

SYSTEM: 16-Channel Electrometer / Picoammeter
Signal Flow: J1 Input -> 16x Filters -> 2x MAX338 Mux -> ADA4530-1 TIA -> ADuCM362 ADC
Range Selection: 4x Relay Ladder (10M/100M/1G/10G) driven by NPN transistors from MCU GPIO

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NET LABEL KEY:
CH_J1A1..16: Input channels from J1 triax connector
MUX_A1..A8, MUX_B1..B8: Filtered signals to mux inputs
TIA_IN: Mux output to TIA inverting input
INV_OUT: TIA feedback nodes (relay ladder)
AIN0: TIA output to MCU ADC
AIN1: VREF monitor to MCU ADC
ADDR_A0..A2, EN_A, EN_B: Mux address and enable
GP1_0..GP1_3: MCU GPIO to relay NPN drivers
RELAY_0..RELAY_3: NPN collector to relay coils

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